

# Hyperjump via the Fourth Spatial Dimension: A Programme Note on Chord Trajectories through the Bulk of a Closed Three-Sphere Universe

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## Abstract

Within the Quaternion-Hypersphere Theory of Spacetime [Nopper 2025a], the spatial universe is a closed three-sphere  $S^3$  of geometric radius  $R_{\text{geom}}(\tau)$  embedded in a four-dimensional Euclidean space with coordinates  $X^A = (\xi, x, y, z)$ , often packaged as the quaternion  $q = \xi + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ . This note examines whether two points of  $S^3$  separated by an angular distance  $\theta$  can be connected by a trajectory that leaves the hypersurface, traverses the interior of the embedding  $\mathbb{R}^4$  as a straight chord, and re-enters at the destination. The chord is strictly shorter than any geodesic arc on  $S^3$ ; in operational terms — for an observer confined to  $S^3$  — the traveller arrives sooner than a light signal travelling along  $S^3$  would, so the trajectory is superluminal in  $S^3$  projection while remaining timelike in the bulk metric. As a candidate dynamical skeleton we propose a separate quaternionic linear sigma model with order-parameter field  $\phi^A(X)$  and vacuum expectation value  $v$ ; the intended physical brane is the locus near  $|X| = R_{\text{geom}}$  where Standard-Model zero modes are localized by the background profile of  $\phi^A$ . The resulting field equations are globally smooth on  $\mathbb{R}_\tau \times \mathbb{R}^4$ , including at  $\phi^A = 0$ , and chord traversal becomes either a classical over-the-barrier excitation of the localization field or a quantum-tunnelling process. Eight substantial pieces remain to be supplied — cosmic expansion, coupling to gravity, Standard-Model matter content, the Goldstone-mode mass problem, a fixed value for the coupling  $\lambda$  and vev  $v$ , reconciliation with the intrinsic light-speed bound of [Nopper 2025a], a field-theoretic tunnelling-rate calculation, and a UV completion of the (non-renormalizable) quartic interaction. These open subproblems are listed explicitly with first-pass closures. The note is therefore best read as a *programme* rather than a finished theory; it makes no unique precision predictions for laboratory observables beyond those already implied by the underlying braneworld and sigma-model literature, and there is no experimental evidence for macroscopic traversal of extra dimensions.

**Keywords:** braneworld cosmology; linear sigma model; quaternion-hypersphere spacetime; extra dimensions; superluminal trajectories; Coleman bounce.

## 1 Introduction

**Status and limitations.** This document is a *programme note*, not a finished physical theory. It assumes a closed  $S^3$  spatial topology (a premise of [Nopper 2025a], not established by observation), it does not construct a self-consistent 5D gravity + Standard-Model-localization solution, and it does not contain a chronology-protection proof that the bulk-light-cone reformulation in Subproblem 6 is free of closed timelike curves. The “first-pass closures” given for the eight open subproblems are pointers to the appropriate textbook tools, *not* completed calculations for this specific setup. Readers should treat all “in-principle possible” statements as conditional on those three items.

The possibility of connecting widely separated points of the observable universe by a higher-

dimensional shortcut is a recurring theme of speculative physics, from the Einstein–Rosen bridge and traversable wormholes [Morris & Thorne 1988] to the Alcubierre warp metric [Alcubierre 1994]. These constructions invariably require either non-trivial topology of the spacetime manifold or sources that violate the standard energy conditions of general relativity. In contrast, the Quaternion-Hypersphere Theory of Spacetime [Nopper 2025a] already comes equipped with a natural higher-dimensional embedding: the spatial universe is taken to be a closed three-sphere  $S^3$  sitting inside an auxiliary four-dimensional Euclidean space  $\mathbb{R}^4$ . The present note explores the consequences of taking that embedding seriously as a physical arena rather than as a purely auxiliary mathematical construct.

The paper is organized as follows. Section 2 establishes the elementary geometric statement that a straight chord through  $\mathbb{R}^4$  is strictly shorter than any geodesic arc on  $S^3$  between the same two points. Section 3 introduces a quaternionic linear sigma model as a candidate dynamical skeleton on the bulk and identifies the parametric scales it generates. Section 4 enumerates eight open subproblems that must be closed before the proposal can be considered a candidate physical theory, with first-pass closures pointing to the appropriate textbook tools. Section 5 outlines a tiered experimental programme and explicit falsification criteria. Section 6 comments on the relationship to the interval-number framework of [Nopper 2025b]. Section 7 summarizes the status of the proposal.

## 2 The Geometric Argument

A point on the spatial hypersphere satisfies  $|X| = R_{\text{geom}}$ . (Note: that the spatial universe is closed with  $S^3$  topology is a *premise* taken from [Nopper 2025a]; current cosmological data are consistent with this premise but do not establish it — see §Experimental Programme.) Two points  $X_1, X_2 \in S^3_{R_{\text{geom}}}$  separated by angular distance  $\theta$  admit two path types: a geodesic *arc* of length  $R_{\text{geom}}\theta$  that stays on  $S^3$ , and a straight-line *chord* through the embedding  $\mathbb{R}^4$  of length  $2R_{\text{geom}}\sin(\theta/2)$ . The two-dimensional analogue ( $S^1 \subset \mathbb{R}^2$ ) is shown in Figure 1.

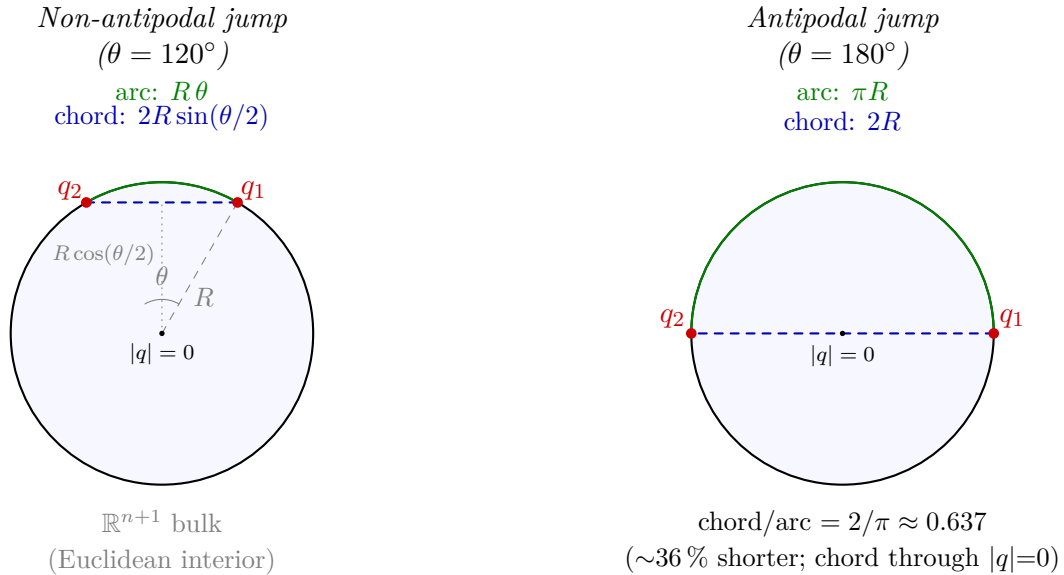


Figure 1: Chord through  $\mathbb{R}^{n+1}$  versus geodesic arc on  $S^n$  (illustrated for  $n = 1$ ). The hyperjump trajectory (dashed blue) leaves  $S^n$ , traverses  $\mathbb{R}^{n+1}$ , and re-enters at  $q_2$ .

| Path                                          | Parameterization                                              | Length                            |
|-----------------------------------------------|---------------------------------------------------------------|-----------------------------------|
| Geodesic on $S^3$ (normal travel)             | $\text{slerp}(X_1, X_2, t)$ , with $ X(t)  = R_{\text{geom}}$ | $R_{\text{geom}} \theta$          |
| Chord through $\mathbb{R}^4$ bulk (hyperjump) | $(1-t)X_1 + tX_2$ , with $ X(t)  \leq R_{\text{geom}}$        | $2R_{\text{geom}} \sin(\theta/2)$ |

Since  $\sin(\theta/2)/(\theta/2) \leq 1$  for all  $\theta \in (0, \pi]$ , the chord is never longer than the arc. The maximum shortcut occurs at the antipodal point ( $\theta = \pi$ ): chord  $= 2R_{\text{geom}}$  versus arc  $= \pi R_{\text{geom}}$  — a reduction of approximately 36%. At bulk speed  $u \leq c$  the crossing time is  $\Delta\tau = 2R_{\text{geom}} \sin(\theta/2)/u$ , strictly finite and timelike with respect to the bulk metric.

It must be acknowledged that, from the perspective of an  $S^3$ -confined observer, the chord traverser arrives sooner than a light signal travelling along  $S^3$  between the same endpoints. The effective  $S^3$ -projected speed for an antipodal chord at bulk speed  $c$  is  $\pi c/2 \approx 1.57c$ . The trajectory is therefore operationally superluminal in  $S^3$  projection. This is consistent with the bulk light-cone but in tension with the FTL prohibition in [Nopper 2025a, §Light-Speed Bound], which is formulated *intrinsically* on  $S^3$ . The resolution adopted here — extending the causal-cone definition to the bulk Minkowski metric — is given as the first-pass closure of subproblem 6 below.

For an antipodal jump the chord passes through  $|X| = 0$  at its midpoint; for non-antipodal jumps it passes through a minimum norm  $R_{\text{geom}} \cos(\theta/2) > 0$ . The bulk origin  $|X| = 0$  is a point of the auxiliary embedding  $\mathbb{R}^4$  at the present cosmic time  $\tau$ ; it is *not* the FLRW initial singularity at  $\tau = 0$ , although both share the property that the geometric radius vanishes.

### 3 A Candidate Dynamical Skeleton: Quaternionic Linear Sigma Model

The SpacetimeTheory designates  $\mathbb{R}^4$  as an *auxiliary* embedding space and defines matter only on  $S^3$  [Nopper 2025a, §Foundations]. To make the hyperjump a physical process, the embedding coordinate  $X^A$  must be supplemented by independent dynamics: an action principle for fields living on the bulk that admits both  $S^3$ -localized ground states and chord-like excited trajectories.

We propose, as the simplest candidate skeleton, a Lorentz-invariant linear sigma model with global  $O(4)$  symmetry on the bulk  $\mathcal{B} = \mathbb{R}_\tau \times \mathbb{R}^4$ . The dynamical field is a real 4-component order parameter  $\phi^A(X)$  ( $A = 1, \dots, 4$ ), not the embedding coordinate  $X^A$  itself:

$$\mathcal{L}_\phi = -\frac{1}{2} \eta^{MN} \partial_M \phi^A \partial_N \phi^A - \lambda (\phi^A \phi^A - v^2)^2 \quad (1)$$

with five-dimensional Minkowski signature  $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$  in natural units ( $\hbar = c = 1$ ) and implicit sums over  $M, N$  and  $A$ . Structurally this is the Higgs sector with the field-space vacuum manifold  $\{\phi^A \phi^A = v^2\} \cong S^3$ . The physical spatial hypersphere remains the geometric locus  $|X| = R_{\text{geom}}$ ; identifying that locus with the low-energy brane requires a background profile  $\phi_0^A(r)$ , where  $r = |X| - R_{\text{geom}}$ , that localizes Standard-Model zero modes near  $r = 0$ . We emphasize that such a radial domain-wall profile at a specific finite radius  $|X| = R_{\text{geom}}$  does *not* form spontaneously in flat  $\mathbb{R}^4$  from the potential  $\lambda(|\phi|^2 - v^2)^2$  alone — the natural codimension-four defect of an  $O(4)$ -symmetric model is a global-monopole hedgehog  $\phi^A \propto \hat{X}^A$  centred at  $|X| = 0$ , not a thin shell at finite radius. Pinning the brane at  $|X| = R_{\text{geom}}(\tau)$  requires the additional dynamical input of Subproblems 1 and 2 (cosmic-expansion sector and coupling to gravity);  $\phi_0^A(r)$  should therefore be read as the ansatz one wishes a self-consistent solution to produce, not as an established solution of the present action.

**Conventions and dimensions.** Throughout this note we use natural units  $\hbar = c = 1$  unless otherwise stated. In five spacetime dimensions a canonically normalized real scalar  $\phi^A$  has mass

dimension  $3/2$ ; the vev  $v$  has mass dimension  $3/2$ ; and  $\lambda$  has mass dimension  $-1$  (confirming non-renormalizability — see Subproblem 8). The geometric radius  $R_{\text{geom}}$  has dimension of length and is not the same object as  $v$ . A complete gravity/localization solution must determine how  $v$ ,  $\lambda$ , and the background profile  $\phi_0^A(r)$  set the brane thickness and the observed geometric radius.

The skeleton does three things:

| Feature                                         | Mechanism                                                                                                                 |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Brane localization becomes dynamical            | Matter confinement is controlled by $\phi_0^A(r)$ rather than imposed as the kinematic constraint $ X  = R_{\text{geom}}$ |
| Equations of motion are globally smooth         | $\square\phi^A = 4\lambda( \phi ^2 - v^2)\phi^A$ is well-posed everywhere on the bulk, including at $\phi^A = 0$          |
| The central field configuration is non-singular | $V(0) = \lambda v^4$ is finite; leaving the brane encounters a finite barrier, not a divergence                           |

Excitations with energy density above the central barrier  $\rho_\star = \lambda v^4$  can in principle push localized states into the bulk along chord trajectories; excitations below the barrier tunnel quantum-mechanically. The semiclassical estimate must, however, be adapted to the symmetry of the potential:  $V(\phi) = \lambda(\phi^A\phi_A - v^2)^2$  has two *degenerate* minima on the vacuum manifold  $|\phi| = v$ , so the standard Coleman false-vacuum bounce [Coleman 1977] does not apply — there is no finite-action  $O(d)$ -symmetric instanton that starts at  $|\phi| = v$ , passes through  $|\phi| = 0$ , and returns. The correct saddle is instead the planar domain wall (kink) of surface tension

$$\sigma = \frac{4\sqrt{2}}{3} \sqrt{\lambda} v^3, \quad (2)$$

and the Euclidean action for a cargo of transverse 3-area  $A_c$  traversing a chord of length  $\ell_{\text{chord}} = 2R_{\text{geom}} \sin(\theta/2)$  is dominated by the worldvolume of the wall-tube wrapping that cargo,

$$B_{\text{phys}} \sim \sigma (P_c \ell_{\text{chord}} + 2 A_c), \quad (3)$$

with  $P_c$  the 2-perimeter of the cargo transverse to the chord. Once gravity is included, the appropriate extension of (3) is a Coleman–De Luccia-type calculation [Coleman & De Luccia 1980] for the wall worldvolume on the curved bulk, not for a Coleman bounce. The parameters  $\lambda$  and  $v$  then control the *brane thickness*  $\ell_\perp \sim 1/(\sqrt{\lambda}v)$ , the *barrier energy density*  $\rho_\star = \lambda v^4$ , the *wall tension* (2), and via (3) the parametric *action cost* of transporting an object through the bulk.

This much is a clean dynamical skeleton. It is **not yet a complete physical theory**.

## 4 Open Subproblems

The skeleton above leaves eight substantial pieces unsupplied. Each is an explicit open problem; the model becomes a candidate physical theory only when all are addressed. Each item is given a textbook-level *first-pass closure* inline — a starting point, not a final solution. The depth of the residual work varies considerably: items 1–5 reduce to standard exercises in scalar-field cosmology, braneworld gravity, brane localization, gauge symmetry breaking, and dimensional analysis; items 6–7 require dedicated technical follow-ups (causal-structure reformulation, full bounce computation); item 8 is a deferral to a UV-completion programme that lies outside the scope of any single note.

1. **Cosmic expansion.** [Nopper 2025a] has  $R_{\text{geom}} = R_{\text{geom}}(\tau)$  (FLRW). The static brane location in the skeleton is incompatible. *First-pass closure:* introduce an additional collective coordinate

$\Phi(\tau)$  controlling the equilibrium brane radius and a potential  $U(\Phi)$  with a kinetic term  $-\frac{1}{2}(\partial\Phi)^2$ . (We emphasise:  $\Phi$  is an effective field controlling the brane location in the embedding  $\mathbb{R}^4$ ; it is *not* the FLRW scale factor itself, which must instead emerge from the induced 4D metric of Subproblem 2.) The homogeneous mode then satisfies  $\ddot{\Phi} + 3H\dot{\Phi} + U'(\Phi) = 0$ , the standard scalar-field-cosmology equation [Linde 1990]; for slow-roll choices of  $U$  this reproduces de Sitter expansion. Full closure requires coupling to 5D gravity (Subproblem 2) so that  $H$ , the relation between  $\Phi$  and the sigma-model profile, and the resulting FLRW scale factor are all solutions rather than postulates.

2. **Coupling to gravity.** The bulk is treated as flat. To recover the FLRW geometry as a *solution* rather than a postulate, the action must be coupled to 5D General Relativity,

$$S = \int d^5X \sqrt{-g} \left[ \frac{\mathcal{R}^{(5)}}{2\kappa_5^2} + \mathcal{L}_\phi \right], \quad (4)$$

in the spirit of Randall–Sundrum-type braneworlds. *First-pass closure:* in the thin-brane limit  $\ell_\perp \rightarrow 0$ , the energy-momentum tensor of the localized  $\phi_0^A(r)$  background degenerates to a delta-function shell source near  $|X| = R_{\text{geom}}$  and the 5D Einstein equations reduce to Israel junction conditions [Israel 1966]; the induced 4D geometry is then standard FLRW with the brane tension  $\sigma = \int dr V(\phi_0)$  playing the role of an effective cosmological constant. A treatment with finite  $\ell_\perp$  requires solving the coupled  $(g_{MN}, \phi^A)$  system numerically.

3. **Standard-Model matter on the brane.** Real matter is not a quaternion sigma model. *First-pass closure:* introduce a 5D Dirac fermion  $\Psi$  with a Yukawa mass profile  $m_\Psi(r) = hf[\phi_0(r)]$  chosen so that  $m_\Psi(r)$  changes sign across the brane at  $r = |X| - R_{\text{geom}} = 0$  [Rubakov–Shaposhnikov 1983]. The transverse Dirac equation  $[i\gamma^\perp \partial_r - m_\Psi(r)]\Psi = 0$  then admits a normalizable zero mode with profile  $\Psi_0(r) \propto \exp[-\int^r m_\Psi(r') dr']$  peaked on  $S^3$  and exponentially suppressed in the bulk; this is the 4D Standard-Model fermion. Gauge bosons can be localized by analogous dilaton-type couplings [Dvali–Shifman 1997]. Recovering the full SM spectrum (chirality, generations, Higgs) within this framework is non-trivial but follows established braneworld phenomenology.
4. **Goldstone-mode mass problem.** The breaking  $O(4) \rightarrow O(3)$  in field space produces three exactly massless scalars not seen in nature. *First-pass closure:* gauge the connected component  $SO(4)$  by introducing gauge fields  $A_M^a$  ( $a = 1, \dots, 6$ , since  $\dim SO(4) = 6$ ) and replacing the partial derivative by the covariant derivative,

$$\partial_M \phi^A \longrightarrow D_M \phi^A = \partial_M \phi^A - ig A_M^a (T^a)^A{}_B \phi^B, \quad (5)$$

plus a Yang–Mills kinetic term  $-\frac{1}{4}F_{MN}^a F^{aMN}$ . After symmetry breaking, the three broken generators give three massive vector bosons with masses  $m_A \sim gv$ ; the three Goldstones become their longitudinal components via the Higgs mechanism. The remaining  $SO(3)$  gauge bosons stay massless and would need to be either confined or identified with a known sector — itself a non-trivial phenomenological constraint.

5. **Determination of  $\lambda$  and  $v$ .** The coupling and vev are unconstrained, so the model is not predictive. *First-pass closure:* in the present setup the transverse direction (radial in the embedding  $\mathbb{R}^4$ , across the  $S^3$  brane) is *non-compact*, so  $\ell_\perp \sim 1/(\sqrt{\lambda}v)$  is a brane-thickness / warping length, not a Kaluza–Klein compactification radius. Current laboratory bounds on deviations from the Newtonian inverse-square law constrain a new transverse length scale to  $\ell_\perp \lesssim 52 \mu\text{m}$  at 95 % C.L. [Lee et al. 2020], superseding the earlier  $\sim 44 \mu\text{m}$  bound reviewed in [Adelberger et al. 2003]; this translates into  $\sqrt{\lambda}v \gtrsim 1/\ell_{\text{exp}} \sim 4 \text{ meV}$  at the present level of precision. This constrains only the product  $\sqrt{\lambda}v$ ; pinning unique values requires an independent observable (e.g. a continuum graviton signature in a non-compact transverse direction, a tunnelling rate, a radial-mode mass, or a cosmological constraint on  $\Phi$ ). The

precise translation of the laboratory bound into a constraint on  $\sqrt{\lambda}v$  depends on the unsolved gravity sector (Subproblem 2) and is therefore itself only first-pass.

6. **Operational FTL versus [Nopper 2025a]’s prohibition.** Chord traversal is timelike in the bulk but superluminal in  $S^3$  projection. *First-pass closure:* adopt the bulk Minkowski light-cone of  $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$  (natural units) as the *fundamental* causal structure, and reinterpret [Nopper 2025a]’s intrinsic light-speed bound as an emergent statement about  $S^3$ -confined matter — strictly valid only for trajectories that remain on the brane. Chord trajectories then satisfy the bulk light-cone. The standard “FTL implies CTCs by Lorentz-boosting” argument does *not* directly apply here, because the FLRW embedding picks out a preferred cosmic-time slicing and brane Lorentz boosts are not a true symmetry of the full theory. *However*, this observation alone does not constitute a chronology-protection proof: brane-confined observers in different regions of  $S^3$  enjoy *approximate* local Lorentz invariance, and one must verify explicitly that no concatenation of (i) brane-confined sub-light trajectories and (ii) allowed bulk chords can close into a causal loop with respect to the bulk light-cone. We do not supply that verification here; it is an explicit open subproblem of this note. The cost is conceptual: the “intrinsic” formulation of light-speed in [Nopper 2025a] must be downgraded to an effective brane-projected bound, and Lorentz invariance of the brane physics becomes approximate rather than exact (constrained by precision tests of vacuum Lorentz invariance — see Tier 1 below). An alternative — adding a chronology-protection term that suppresses access to  $\phi^A \approx 0$  — preserves the intrinsic formulation but reintroduces an arbitrary suppression scale.

7. **Tunnelling rate.** A field-theoretic semiclassical calculation should replace the 1D-WKB heuristic. *First-pass closure:* the symmetric double-well  $V(\phi) = \lambda(\phi^A \phi^A - v^2)^2$  has two *degenerate* vacua on  $|\phi| = v$ , so a Coleman-type  $O(d)$ -symmetric bounce [Coleman 1977] interpolating between  $|\phi| = v$  at infinity and  $|\phi| = 0$  at the centre does not exist as a finite-action saddle — the relevant process is not false-vacuum decay. The appropriate saddle is the planar kink with surface tension  $\sigma = \frac{4\sqrt{2}}{3}\sqrt{\lambda}v^3$  (Eq. 2); the Euclidean action for a cargo traversing a chord of length  $\ell_{\text{chord}}$  is dominated by the wall-tube worldvolume,

$$B_{\text{phys}} \sim \sigma(P_c \ell_{\text{chord}} + 2A_c),$$

giving  $\Gamma/\mathcal{V}_5 \sim \mu^5 e^{-B_{\text{phys}}/\hbar}$ . Substituting the laboratory-saturating value  $\sqrt{\lambda}v \sim 1/\ell_{\perp} \sim 4 \text{ meV}$  from Subproblem 5 and any macroscopic cargo (e.g. a 1 m-radius object across a 1 ly chord) gives  $B_{\text{phys}} \gtrsim 10^{30}$ , so unaided hyperjumps are exponentially suppressed beyond microscopic probes and would require an external coherent source to occur on macroscopic timescales. Once gravity is included, the appropriate extension is a Coleman–De Luccia-type computation [Coleman & De Luccia 1980] of the wall worldvolume on the curved bulk; this fixes the exponent’s  $O(1)$  coefficient and the prefactor. A companion numerical implementation (`numerics/bounce.py` in this repository) verifies  $\sigma$  analytically and numerically and tabulates  $B_{\text{phys}}$  for sample cargos.

8. **UV completion.** The quartic interaction  $\lambda(\phi^A \phi^A - v^2)^2$  is non-renormalizable in five spacetime dimensions; the Lagrangian is therefore at best an effective theory below some cutoff  $\Lambda$ . *First-pass framing:* treat the action as a Wilsonian EFT valid for energies  $E \ll \Lambda \lesssim M_{\text{Pl}}^{(5)}$  (the 5D Planck mass after Subproblem 2 is implemented). Within that range the model is predictive in the EFT sense — operators of dimension  $> 5$  are suppressed by powers of  $E/\Lambda$  — which suffices for any prediction at sub-Planckian scales, including the chord-traversal energetics. A genuine UV completion (asymptotic safety, embedding in a higher-symmetry or string-theoretic theory) lies beyond the scope of this note and is not supplied here.

A consistent closure of items 1–2 alone would already promote the proposal from a kinematic skeleton to a candidate dynamical theory; items 3–5 and 7 would make it phenomenologically

meaningful; item 6 reframes the causal structure of [Nopper 2025a]; item 8 is a deferral to a UV-completion programme.

## 5 Experimental Programme

The model’s predictions split cleanly into two regimes: properties of the *underlying sigma-model and brane structure*, which are testable with existing or near-term experiments, and properties of *macroscopic chord traversal itself*, which are not testable with any foreseeable technology.

### 5.1 Tier 1 — Tests parasitic on running or near-term experiments

| Observable                                                 | What it constrains                                                                          | Status                                                                                                                                                                               |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sub-mm gravity (Eöt-Wash and successors)                   | Brane thickness $\ell_\perp$ constrains $\sqrt{\lambda}v$ via Subproblem 5                  | Currently $\ell_\perp \lesssim 52\,\mu\text{m}$ (95 % C.L.) [Lee et al. 2020], superseding $\sim 44\,\mu\text{m}$ [Adelberger et al. 2003]; next-generation aims at the micron scale |
| Fifth-force / new long-range vector searches               | Residual unbroken $SO(3)$ gauge bosons (Subproblem 4) must be confined or below sensitivity | Tight bounds on any massless vector mediator                                                                                                                                         |
| Vacuum Lorentz-invariance / preferred-frame tests          | Approximate brane Lorentz invariance required by Subproblem 6                               | Existing bounds require any bulk-frame effects to be strongly suppressed                                                                                                             |
| CMB scalar/tensor ratio (Planck, LiteBIRD, CMB-S4)         | Shape of the inflaton potential $U(\Phi)$ in Subproblem 1                                   | Tightening $r$ disfavors specific $U(\Phi)$                                                                                                                                          |
| Spatial-curvature and topology tests (CMB matched circles) | The model presupposes a closed three-sphere universe of finite radius $R_{\text{geom}}$     | Data consistent with flat or slightly closed; no positive detection                                                                                                                  |

### 5.2 Tier 2 — Dedicated or next-generation facilities

| Observable                                                      | What it constrains                                                          |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------|
| KK-like resonances and missing-energy events at HL-LHC / FCC-hh | Bulk-propagating excitations of $\phi^A$ ; matter briefly entering the bulk |
| Precision Higgs couplings                                       | Mixing between the SM Higgs and the radial (massive) mode of $\phi^A$       |
| Dark-energy equation of state $w(z)$                            | Trajectory of $\Phi(\tau)$ on $U(\Phi)$                                     |

### 5.3 Tier 3 — The hyperjump itself

Given the exponential suppression in Subproblem 7, no terrestrial experiment can plausibly produce a macroscopic chord traversal. Conceptual milestones, in increasing order of difficulty:

1. **Microscopic tunnelling event.** Observation of a single quantum appearing at an unexpected location with a rate consistent with the bounce calculation, after excluding all conventional channels. Astronomically rare for any currently allowed  $(\lambda, v, \phi_0)$  background.
2. **Coherent-source proof of principle.** A configuration raising  $\phi^A$  above the central barrier in a small region. The required energy density  $\rho_\star = \lambda v^4$  is bounded below indirectly through Subproblem 5’s lower bound on  $\sqrt{\lambda}v$ , and for any macroscopic cargo is extreme by any

laboratory standard — qualitatively comparable to the absolute energy-scale problem in warp/wormhole proposals, though here with positive sign.

3. **Macroscopic chord traversal.** Direct observation of an object appearing near its antipode in less than light-travel time on  $S^3$ . Not forbidden by the framework, but on no plausible engineering horizon.

## 5.4 Falsification criteria

The proposal is *falsified* (in the strong Popperian sense) if any of the following obtains:

- Cosmological curvature and topology constraints drive the allowed parameter interval entirely outside the closed three-sphere geometry required by [Nopper 2025a];
- Sub-mm gravity, fifth-force, and localization constraints leave no overlapping parameter region for  $\ell_\perp$ ,  $\lambda$ ,  $v$ , and the Standard-Model zero-mode profiles;
- Precision tests of Lorentz invariance in vacuum exclude the bulk light-cone reformulation of Subproblem 6 at the level required for the chord trajectories to remain consistent with observed brane physics;
- A graviton spectrum inconsistent with a single non-compact transverse dimension — for example, a sharp Kaluza–Klein tower with discrete mass spacing characteristic of a *compact* extra dimension, rather than the localized zero mode plus continuum expected for a non-compact transverse direction in the present setup — is established at any future collider or astrophysical observation.

## 5.5 Honest summary

The *theory* is empirically engageable through existing programmes; Tier 1 can strongly constrain it and could potentially falsify specific parameter regions. The *device* — hyperjump propulsion — is not on any plausible engineering horizon and would require physics or technology beyond what is currently available.

## 5.6 In-principle possibility

Subject to the obvious disclaimers, the framework contains no in-principle prohibition against constructing a hyperjump device. What it contains are quantitative obstacles: depositing energy density  $\rho_\star = \lambda v^4$  coherently in the cargo volume, maintaining quantum coherence of the bulk excitation across the chord, and either accepting the exponentially small spontaneous tunnelling rate or engineering a resonant/stimulated enhancement. Each is severe; none is forbidden by the present skeleton.

In this respect the proposal sits in the same epistemic category as traversable wormholes [Morris & Thorne 1988] and the Alcubierre warp metric [Alcubierre 1994]: physically permitted within a self-consistent theoretical framework, but with engineering requirements far beyond present extrapolation. One distinguishing structural feature is worth noting: the chord traversal **does not require violation of the energy conditions of general relativity** — it requires positive-energy excitations of the sigma-model field above a finite barrier  $V(0) = \lambda v^4$ , whereas wormhole and Alcubierre constructions require negative energy density (exotic matter). Whether this counts as a *practical* advantage is debatable, since the absolute energy-density scale required here is itself extreme; what is unambiguous is that the *qualitative* obstacle is different.

The honest classification is therefore: *in-principle possible, technologically remote, and structurally distinct from negative-energy-density proposals*.



## 6 Note on the Interval-Number Framework

Earlier formulations of this proposal invoked the interval-number algebra of [Nopper 2025b] to regularize formal expressions of the type  $\rho_{\text{bulk}} \cdot V_{\text{bulk}} \sim |X|^{-3} \cdot |X|^3 \rightarrow 0 \cdot \infty$  that arise if one extends FLRW-style densities into the bulk via the geometric radius  $|X|$ . Within the sigma-model skeleton above, no such densities appear: the action is governed by the Lagrangian  $\mathcal{L}_\phi$ , which is finite everywhere. The interval-number framework therefore plays no role in the present formulation. It remains an independent, self-consistent algebraic system [Nopper 2025b] and may still be useful for other extensions of the theory in which genuinely indeterminate forms appear.

## 7 Summary

| Framework                                    | Role                                                                                                                                                                        |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quaternion-Hypersphere Theory [Nopper 2025a] | Identifies $\mathbb{R}^4$ as the embedding space and supplies the chord-path geometry of the closed $S^3$ universe                                                          |
| Quaternionic linear sigma model (this note)  | Provides a candidate bulk Lagrangian for an order-parameter field $\phi^A$ whose background profile localizes matter near $S^3$ and whose excited states can enter the bulk |

The model is **geometrically coherent and singularity-free**, but **not yet dynamically complete**: the eight open subproblems listed above must be closed before it can be considered a candidate physical theory. Each admits a textbook-level first-pass closure sketched inline (with item 8 framed as a Wilsonian-EFT deferral), but full closure of any one item is itself a non-trivial research programme. It introduces, beyond [Nopper 2025a], the  $O(4)$ -symmetric quartic potential  $V(\phi) = \lambda(\phi^A \phi^A - v^2)^2$  with coupling  $\lambda$  and vev  $v$ . These parameters have clean physical interpretations (barrier energy density  $\rho_\star = \lambda v^4$ , brane thickness  $\ell_\perp \sim 1/(\sqrt{\lambda} v)$ , transit-energy cost  $\rho_\star V_{\text{cargo}} \ell_{\text{chord}}$ ) but no fixed numerical values at present. The model makes no unique precision predictions yet, and there is no experimental evidence for macroscopic bulk traversal.

## References

### Foundations of this work

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### Extra dimensions and braneworld models

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### Speculative spacetime engineering (for comparison)

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### Background (Wikipedia)

- [N-sphere](#)
- [Quaternion](#)
- [Linear sigma model](#)
- [Brane cosmology](#)
- [Kaluza–Klein theory](#)
- [Faster-than-light](#)
- [Geodesic](#)

## Inspirational background (science fiction)

The motivating question of this note — whether a higher-dimensional shortcut between distant points of our spatial universe could be physically realisable — is a recurring motif in twentieth-century science fiction. The works listed below served only as *inspiration* for posing the question in the first place; they form no part of the physics or mathematics presented above, and no claim is made that the corresponding fictional devices realise the proposed mechanism.

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- [Herbert 1965] Herbert, F. *Dune*. Chilton Books, 1965. — foldspace navigation.
- [Roddenberry 1966] Roddenberry, G. *Star Trek* (TV series). NBC, 1966. — warp drive through subspace.
- [Lucas 1977] Lucas, G. *Star Wars: Episode IV – A New Hope*. 20th Century Fox, 1977. — hyperspace.
- [Banks 1987] Banks, I. M. *Consider Phlebas*. Macmillan, 1987. — discrete jump drives.